

INTRODUCTION:

People who need better tools for developing new systems, solving problems, and selecting solutions include:

- Those doing research and development in technical and nontechnical fields
- Technology managers and other managers
- Those who solve problems

A new model for problem solving. The model is based on the theory of problem solving called TRIZ. TRIZ is showing up more and more frequently today in success stories of innovative solutions to problems in technical and in nontechnical fields.

TRIZ comes from the Russian phrase *teoriya rezheniya izobretatelskih zadach*. The English translation is the “theory of inventive problem solving.” TRIZ has its origins in the former Soviet Union, where it was founded by Genrich Saulovich Altshuller (1926–1998). In research on patents that started at the end of World War II, Altshuller found that a variety of different engineering systems and technologies had common patterns of evolution. The inventor (or any problem solver) can learn these patterns of evolution, use them to develop new technology consciously and systematically, and avoid many fruitless trials and errors.

Why do we need a new theory? Without a theory, people generate ideas by guesswork and then select the ones they like or those they think other people will like. With TRIZ, you will be able to generate better ideas faster, and you will have a basis for selecting the best ideas, the ideas that will solve your problem effectively and form a basis for further improvements. People cannot select ideas properly and cannot produce better ideas effectively if they are unaware of the common features of good solutions: resolving contradictions and making use of idle resources.

3.1 COMMON FEATURES OF GOOD SOLUTION

Good solutions have several common features. The good idea does the following:

- Resolves contradictions
- Increases the “ideality” of the system
- Uses idle, easily available resources

In addition to their everyday meanings, these words have specific technical meanings in TRIZ. By working with the TRIZ concepts, you will learn to apply them to your problems, to develop good solutions, and to select the best solutions from all that are proposed.

Three Basic Concepts for Reaching the Best Solution

1. A good solution resolves the contradiction that is the cause of the problem. There are two kinds of contradictions:
 - a) “Tradeoff contradiction” means that if something good happens, something bad happens, too.
 - b) “Inherent contradiction” means that I want that one thing that has two opposite properties.
2. The “ideality” of a system is the measure of how close it is to the perfect system. The perfect system (called the “ideal final result” in TRIZ) has all the benefits the customer wants, at no cost, with no harmful effects. So, a system increases ideality when it gives you more of what you want or less of what you do not want, does it at a lower cost, and usually with less complexity.
3. Unseen, idle resources of the system are used to reach these seemingly incompatible goals. These resources include energy, materials, objects, information, or things that can be made easily from the resources that are in the system or nearby.

3.1.2 RESOLVE CONTRADICTION:

Construct a model for problem solving. The model is like a general map that shows, by words and pictures, how to use the most important TRIZ features in problem solving.

One of the early insights of the TRIZ researchers was that solving a problem meant removing a contradiction. If we compare the TRIZ problem-solving methodology to a tree, the concept of contradiction can be compared to the seed, from which we can grow the whole tree. If we would like to express the idea of TRIZ by a single word, that word would be “contradiction.”

The problem-solving model uses five concepts:

- Contradiction
- Resources
- The ideal final result
- The patterns of evolution
- Innovative principles

Contradiction:

- Solving a problem means removing a contradiction.
- Formulate the contradiction

Resources:

- Resources are available, but idle and often invisible substances, energy, properties, and other things in or near the system can be used to resolve the contradiction.
- Mapping the resources

Ideal final result:

- The ideal final result is achieved when the contradiction is resolved. The desired features should be obtained without compromise.
- Study the difference between good and weak solutions.
- Increasing ideality is one important feature of good solutions

Patterns of evolution:

- Systems evolve according to certain patterns, not accidentally. The patterns can be used many ways to get new ideas and predict the evolution of the system.
- The same patterns are repeated in the evolution of systems. These patterns can be used for the further development of the system
- One can use the patterns of evolution for selecting solutions, finding or solving problems, forecasting evolution, and transferring solutions across industries

Innovative principles:

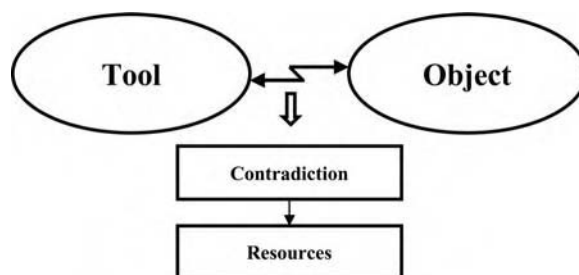
Innovative principles give clues for solving problems. The simplest way to use the principles is to go through them all.

- Some selection criteria are useful for deciding which principles to use for any particular problem. Seek principles that resolve tradeoffs. Try the Contradiction Matrix if it fits your tradeoffs.
- Seek principles that eliminate the inherent contradictions. Resource mapping, the description of the ideal final result, and the patterns of evolution also help to find useful principles.

3.1.3 USE AVAILABLE RESOURCE:

- Sometimes the clear formulation of the contradiction suggests a possible answer to the problem. Usually, however, additional information is needed. Resource analysis helps you find ways to resolve the contradiction.
- Resources are things, information, energy, or properties of the materials that are already in or near the environment of the problem.
- If they can be used directly or modified to make them useful, the problem will appear to have solved itself.
- Think of the resources as the reserves they are invisible at first, because we are accustomed to not seeing them when we look at the problem situation, but we can mobilize these reserves to solve the problem.
- The system should be changed so that the needed improvement seems to appear from nowhere.

- We defined resources as things that are available but are not being used sometimes, we cannot “see” them because we have too many biases.
- Example: We need to change the system so that splitting power is improved, but the ax does not get more difficult to use. The resources in the ax and chunk of wood system are the blade, its edge, form, material, and other properties; the handle and its properties; the chunk of wood and its properties; the surrounding air; and so forth. Depending on how the system is defined, we can also include the person whose arm provides the energy and whose hands provide the transmission of energy to the ax.



1. First, we consider the invisible reserves of systems. All systems have gray zones or proximal zones of development. They are areas where we can find solutions that have potential to be developed, but have not yet been developed. They are zones where business opportunities lie.
2. Second, we look at the benefits from resource analysis. Understanding resources will help you in many ways. The analysis of resources in a situation can independently stimulate new ideas. Understanding resources can resolve the inherent contradiction that creates a problem in the first place. The analysis of resources will help you to foresee the evolution of the system and to understand customer needs that you have not previously identified.
3. Third, we present a simple, handy classification of most useful resources. The tool, object, environment, and macro- and microlevel systems are resources classified by their relationship to the system. On each of the system levels, a variety of resources may be available: substances, energy, space, time, and such. Other resources, such as information, people's skills, and solutions used by other industries are also available.

4. Fourth, we study the seven most important resource groups in detail. The fifth part outlines using resource analysis for explaining undesirable phenomena, for example, occasional faults in products with no visible reason.

In the model of problem-solving resources work as a bridge between the contradiction and the ideal final result. As we have already said, resource mapping can also be used as an independent tool

3.1.4. INCREASE THE IDEALITY

We defined good solutions as those that achieve the ideal final result and resolve the contradiction using idle resources. This is easy to say but hard to do. To make it easier, we need a systematic method for using resources to remove contradictions.

The Law of Increasing Ideality

“The development of all systems proceeds in the direction of increasing the degree of ideality.”

- Systems get simpler, not more complex. The ideal system is the one that has all the useful features and functions of the original system, but has no weight, no volume, requires no labor, no maintenance, consumes no energy, and so on.

“Ideality is the sum of all benefits divided by the sum of all costs and all harm.”

- The maximum value of the equation is reached when the denominator is zero that is how we concluded that the ideal final result is a system that achieves all the benefits with no cost and no harm.
- Because complexity causes an increase in cost and harm, increasing simplicity will increase ideality if the benefits stay the same.
- Ideality is measured by comparing systems. We can easily say which of two alternative systems is closer to ideal in specific circumstances by comparing the ideality equation.
- If benefits increase with no change in cost or harm, then ideality increases. If harm is reduced with no change in benefits or cost, then ideality increases.

Example:

The pocket electronic calculator quickly replaced the slide rule. The calculator was more precise, easier to carry, and easier to remember how to use. When the price of the calculator fell, the slide rule quickly became a museum piece.

3.1.5. TRADE OFF CONTRADICTION:

If a problem exists, clarify the tradeoff behind it. This is the first step in finding the real problem and good solutions.

Tradeoffs are called “technical contradictions,”

We analyze the conflict between two features. These are frequently called “tradeoffs” because the problem solver trades improvement of one feature against decline in another feature in the hope of finding a solution to the problem.

The benefit of TRIZ comes from solving difficult problems and that means resolving tradeoffs. To do this, it is necessary to formulate these tradeoffs or rewrite the problem in a form that makes the tradeoff obvious.

Tradeoff

When something good happens, something bad happens. Alternatively, when something good gets better, something undesirable gets worse.

Some examples of tradeoffs

Contradiction is the core of a challenging problem. Clarify the Tradeoff behind a Problem.

- The product gets stronger (good) but the weight increases (bad).
- Software is made easier to use (good) but versatility decreases (bad)

Why Analyze Tradeoffs?

- TRIZ is a new approach that sees contradictions (tradeoffs and inherent contradictions) as sources of development.
- Resolving the conflicts in a system causes the development of the system.
- Resolving conflicts is the rationale behind successful inventions and innovations.
- If you want to move technology forward, you need to understand the conflicts.

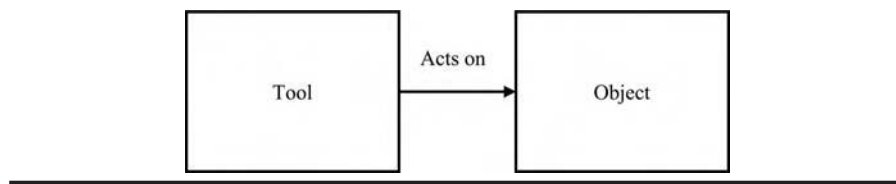
- To do this, one method is to consciously clarify and intensify the conflicts or contradictions.
- Do not treat them as disorders that should be hidden; treat them as important clues to the solution

Defining the Tradeoff:

Tool and Object:

The tradeoff arises in a system consisting of a tool and an object. All engineering systems are built from tools and objects.

- A car is a tool for carrying passengers and cargo.
- A transistor is a tool for switching electrical current on and off
- Knowledge, models, and information are also tools.



The tool is the component that is easiest to change when the problem is resolved.

If the problem is how to decrease the wear of a saw that cuts metal, there are many possibilities change the sharpness or the shape of the teeth, put coatings of tougher metal on the teeth, change the thickness of the blade, add a lubricant during cutting, and so on.

The metal being cut (the object) most often cannot be changed or can be changed very little. The object, on the other hand, gives constraints for the change of the tool. The material being cut limits the choice of cutting tool.

Action means that the tool does something that causes the object to change. The statements of problem and solving the problem become much easier when the tool and the object that are at the heart of the problem are isolated from the numerous other components of the system.

Trade off everywhere

- The system consists of the tool and the object interacting with each other. The tradeoff may appear clearly as a harmful action that accompanies a useful action. There are tradeoffs in the system even when there are no visible harmful effects.
- When the tradeoff is removed, you might think that the perfect system or the ideal final result has been achieved. This achievement, however, is relative and temporary.
- New tradeoffs appear soon, either in the system itself or in other systems that are affected by it.

Example:

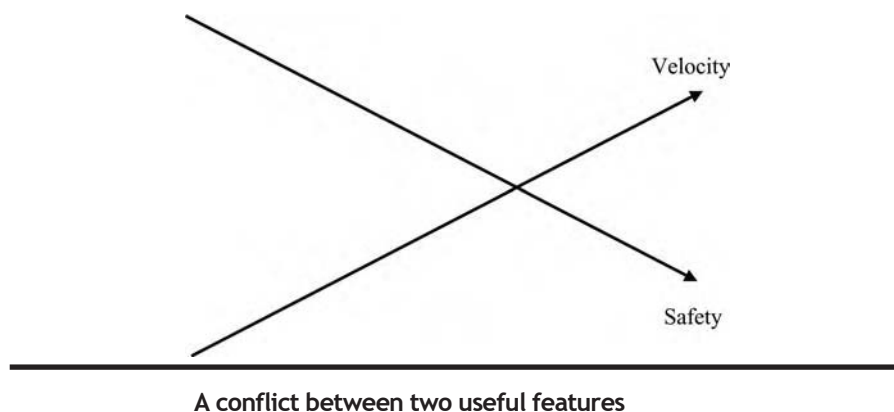
Every new generation of transistors has occupied less space and consumed less energy than the previous one, but the higher density of transistors on the new chips requires more and more complex systems for connection and for heat dissipation.

Features

Features and actions can describe tradeoffs. It is simplest to describe features. Most often, the development of a product means the improvement of the features. Usually, the user does not need the product itself, but needs the features

The Tradeoff between Features

The tradeoff between tool and object shows where the problem lies. It is useful to express the tradeoff as a conflict between two features. When the velocity of the car increases, safety worsens



Where does the tradeoff appear

The problem statement is like the answers to a journalist's questions: who, what, where, when, why, and how. Answering the questions may reveal the tradeoffs. The level of the system is important, too. The system contains some parts of lower level and is itself a part of the higher-level system or macrosystem.

It is useful to formulate problems and tradeoffs on more than one level. The selection of the tradeoff depends on the constraints and conditions of the problem.

How to decide whether to develop the system or remove it

The elimination of a part or operation means, in most cases, that some useful features and functions will disappear. By suggesting the removal of a part of a system or a process, we have created a tradeoff.

If we can formulate the tradeoff explicitly, and then resolve the tradeoff so that the system will have fewer parts and operations and more useful features, we will have significantly improved the system.

Conventional thinking often stops before considering this kind of tradeoff.

What is the new perspective that TRIZ gives here?

Instead of the suggestion: "Suppose we leave this out," we use new questions. What are the good and bad features of this component or operation? What gets better and what gets worse, if we leave this out?" Removing the component or operation is not always possible.

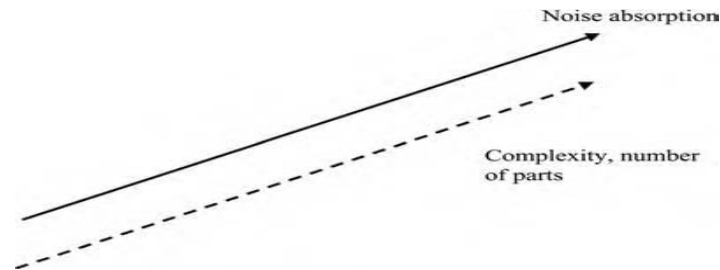
Often, some other solution for the tradeoff is needed, but this method, thinking about what would happen if we removed the component or operation, helps to find the real problem.

From the Problem to the Tradeoff

Let us summarize the points in this chapter. There are five steps for the clarification of a problem:

- Describe pairs of tools and objects and the action that links them. Select one pair. Explain why you picked this tool and object.
- Describe features and conflicts between them.
- Select one tradeoff.
- Explain why you identified this tradeoff.

- Describe the tradeoff graphically and in words.



Constructing the Model of Tradeoffs Using the Lawnmower Example

1.	Describe pairs of tools and objects.	
	a.	Muffler and noise (or air that vibrates)
	b.	Noise and a person (disturbed by noise)
	c.	Engine and exhaust gas (engine moves gas)
2.	Select one pair. Explain why just this tool and object are selected.	
	We select the pair “muffler and noise” because we want to select a limited problem. We want remove the harmful feature with minimal changes in the system.	
3.	Describe features of the selected system of a tool and an object. Describe conflicts in this system.	
	Features: dimensions, weight, absorption capacity, ease of manufacture, form, etc.	
	Conflicts between features:	
	a.	When noise absorption improves (+), the dimensions of a muffler increase (–)
	b.	When noise is suppressed (+), the number of parts increases (muffler needed) (–)
4.	Select one pair of conflicting features. Explain why just this tradeoff is selected. We select the formulation: “When noise is suppressed (+), the number of parts increases (muffler needed) (–).”	
	The reason: A limited problem, very big changes not needed, at the same time the goal is more than simple optimization of the size.	
5.	Describe the tradeoff graphically and in words.	
	In words: When noise absorption capacity improves, dimensions (of the muffler) and number of parts (the muffler is needed) are increased, too. If the system is simplified (or muffler made smaller or removed totally), the noise absorption capacity is lessened or lost.	

3.1.6. INHERENT CONTRADICTION:

The formulation of the inherent contradiction helps to get the best solutions because it helps find the key problem. There are always many problems and tradeoffs, but not all problems are equally important. One problem is key. The solution of this problem leads to the solution of others.

If we can focus on one key problem, we can present problems and their solutions much better in all possible ways.

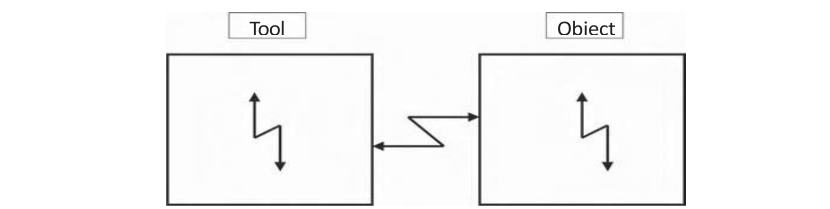
How to Formulate the Inherent Contradiction

Contradictions between a tool and an object or, generally, in the system of the tool and the object.

However, real problems frequently are not simple, and there are conflicts within each object, as well as between them

Examples of typical conflicts

- Heavy-light: Strength, reliability, and safety require that a vehicle should be heavy, but energy economy requires that the same vehicle should be light.
- High energy consumption–low energy consumption: Machinery and tools need to use energy to work on objects, and they should use as little energy as possible.

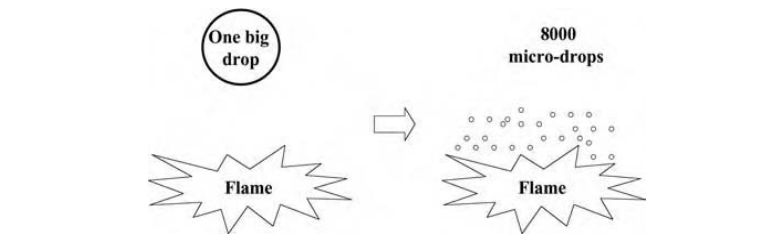


Generally, to improve feature A, an element should have a certain property, but to improve feature B, the same element should have the opposite property.

Using water to fight fires is a technical example that is familiar to everyone. However, the more water used, the more equipment needed. If the fire is put out, large amounts of water can damage the parts of the building that were not damaged by the fire. In some cases, the water does more damage than the fire. The inherent contradiction is “much water–little water.”

Not long ago, a simple and interesting solution was developed. Water is atomized and sprayed in very small micro drops, or mist. Much less water is

needed to extinguish the fire and there is much less damage caused by the water itself. Of course, there is the new contradiction in the complexity of the atomizing equipment. That contradiction is partially eliminated by virtue of decreased water flow and high pressure the size of tubes and other equipment can be decreased. The same volume of water gives 8000 times more drops than a conventional sprinkler and 200 times more than low-pressure mist equipment



The conflict “much water–little water” resolved.

Present and Absent

One important group of inherent contradictions is made up of situations in which the object should be present to get some useful feature and absent to keep the system simple.

Tradeoff: A spare tire in the car increases reliability (+) but occupies space (-).

Inherent contradiction: A spare tire should be present *and* absent. One beautiful solution is to use the normal tire as the spare by making it able to survive a blowout (inherent contradiction solved), rather than making the separate spare tire smaller, as in some cars (compromise).

How to Intensify the Inherent Contradiction

We can sometimes make the internal conflict we just formulated much stronger. This will make the problem seem unsolvable you might think this is a bizarre technique, but it can lead to great solutions.

- The airplane is heavier than air (more dense) yet, in some sense, lighter (thanks to the lifting force).
- The transistor can be a conductor and an insulator at different times.
- Biodegradable materials are present and exist (at certain times) and become absent and nonexistent (at other times).

These examples show how the TRIZ technique of forcing the contradiction to the extreme can be a technique for breakthrough. Analyzing the extreme contradiction helps one break out of conventional thinking into the realm of ideas that get rid of the contradiction and solve the problem.

Reducing Lawnmower Noise: Intensifying Contradiction

Modeling Steps	Example: Reduce Lawnmower Noise
Visible drawback	The lawnmower is too noisy
Tradeoff: the conflict between two feature	When noise absorption improves, the size of muffler and number of parts increase.
Inherent contradiction	Big muffler–small muffler
Intensified inherent contradiction (if it can be intensified)	Big muffler–no muffler

3.2. 30 KEY TRIZ PRINCIPLES:

Understanding the common features of good solutions is crucial to the enhancement of creativity. One obvious conclusion is to simply make a list of the most important features and then use the list for generating bright ideas and successful products.

Good generic solutions across industries have been studied as part of the development of TRIZ.

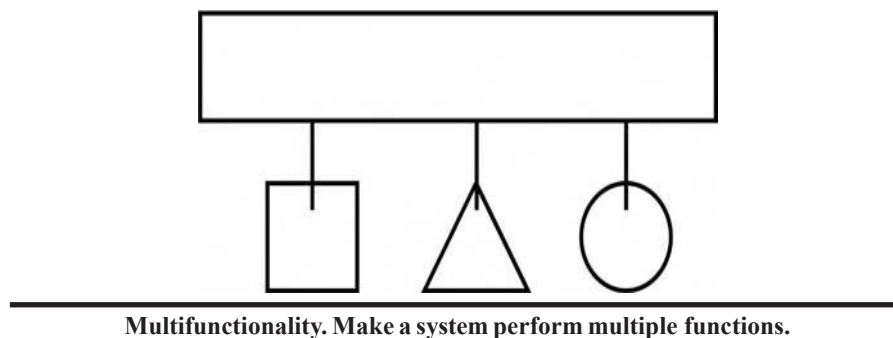
Altshuller and his researchers collected examples of repeated use of the same solutions from patent information. After painstaking work, the information on tens of thousands of good solutions was boiled down to 40 principles in the early 1970s.

The efficiency of the 40 principles increases when they are used together with other tools. Patterns of evolution, ideality, and contradiction analysis may give the same solutions or may give different solutions from the 40 principles considerable overlap is quite common. The tools strengthen and enrich each other.

Problems can be solved and systems improved in different ways, using one principle or using several together. In most solutions, more than one principle is used. When you find an interesting principle, look for other principles that can improve the idea. Often, one principle will give you a concept for a solution, but several may be necessary to get to a practical working solution.

3.2.1. Multifunctionality or universality.

Make a part of an object or system perform multiple functions, and eliminate the need for other parts. number of parts and operations decreases and useful features and functions are retained.



Some examples are

- ABB has developed an electric generator with high voltage. A conventional transformer is not needed because the generator can directly feed the electric network. In some new car designs, the flywheel, alternator, starter, and some other parts are combined into a single component. A single adjustable wrench for all nuts is another example.
- People use the universality principle also. Cross-functional training makes people much less susceptible to layoffs because they have multiple skills instead of one skill. Compare this with the local quality principle.

More examples include:

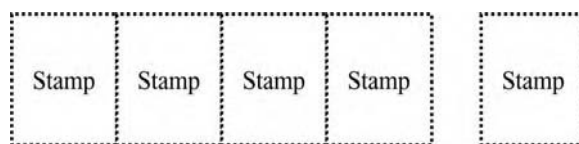
- Team leader acts as recorder and timekeeper.
- One-stop shopping supermarkets sell insurance, banking services, fuel, newspapers, etc.
- Handle of a toothbrush contains toothpaste.

3.2.2. Preliminary action.

Perform, before it is needed, the required change of an object or system (either fully or partially). Prearrange objects so that they can come into action from the most convenient place and without losing time for their delivery.

Examples include:

- Preliminary perforated packaging is easy to open.
- Precut parts for the building of wooden houses save work at the construction site.
- Workers arrange their workspace so that the most frequently used tools (physical, paper, or electronic) are the easiest to reach.
- Do market research, study possible futures, build reserves for changes.
- Prepare solutions to problems that customers are not complaining about today, but may notice tomorrow: environment, moral code.



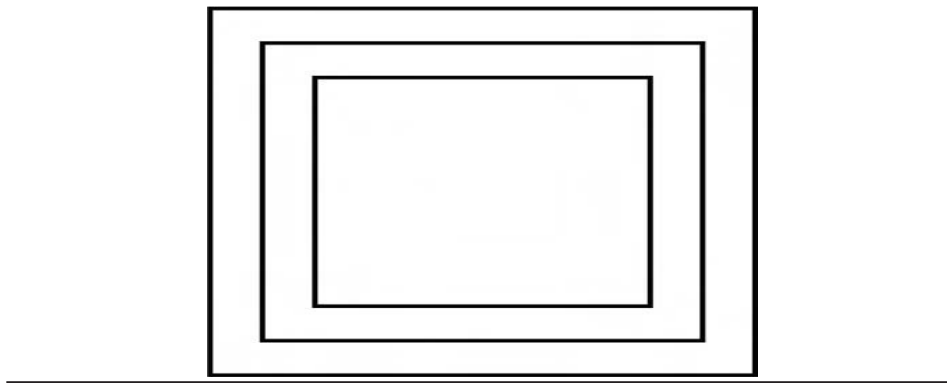
3.2.3. Beforehand compensation.

Prepare emergency means beforehand to compensate for the relatively low reliability of an object or system over time.

- Well-known technological examples are airbags in cars and pressure relief valves in boilers and chemical reactors.
- Nontechnical examples are posting instructions for possible emergency situations fires, the use of narcotics among personnel, environment problems, and preparation of equipment (first aid and rescue kits, fire extinguishers) where they may be needed.
- The frequently asked questions (FAQs) sections of many Web sites are examples of commonly, users are told how to help themselves solve problems that are known to exist in the system.

3.2.4. Nested doll:

Place one object inside another; place each object, in turn, inside the other. Make one part pass through a cavity in the other. The name of this principle comes from the Russian folk-art dolls or *matrushkas*, in which a series of wooden dolls are nested one inside the other



"Nested doll." Place one object inside another.

Examples include:

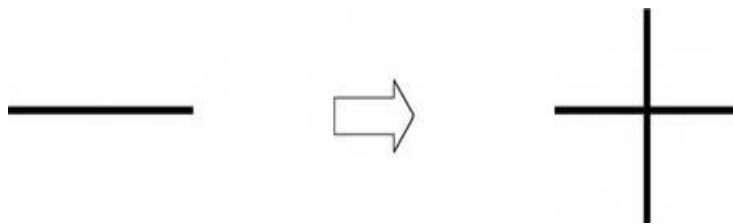
- The double hull in oil tankers
- Telescoping structures (umbrella handles, radio antennas, pointers)
- Business analogies: a special exhibit for one designer inside a boutique store, inside a big market
- File structures in the Windows computer operating system
- Measuring cups or spoons
- Stuffing a turkey with sausage, stuffing the sausage with chestnuts, and so on

3.2.5. BLESSING IN DISGUISE OR “TURN LEMONS INTO LEMONADE.”

Use harmful factors (particularly harmful effects of the environment or surroundings) to achieve a positive effect.

Eliminate the primary harmful action by adding it to another harmful action to resolve the problem.

Amplify a harmful factor to such a degree that it is no longer harmful



“Blessing in disguise.” Use harmful factors to achieve a positive effect.

Some examples are

- Electric charges that are usually harmful can be used for the control of process. Charges can cause fires and explosions, destroy electronic components, make materials stick, and do other harm, but, the same charges, present in most industrial processes, can give valuable information for the optimization of process.
- In coal firing, the measurement of electrical charges on coal dust is used to control burning, decrease harmful nitrogen oxide emissions, and improve efficiency.
- In cement and lime grinding, the product quality can be improved.
- Thermal expansion is often harmful and requires compensating devices, but sometimes, it can be used to make a strong and reliable joint without fixing components.
- In an organization, complaints and destructive critique are negative “charges” that can be modified and used to bring about positive change.
- Singing the blues is an example also. The singer turns personal hardship into entertainment for others.
- Virus attacks in computer networks are never good, but each time the system survives an attack, information is generated that makes the system better protected from the next attack. This is similar to the way the body works surviving an illness generates antibodies that protect the victim from the next incipient infection.
- In many situations, people have turned lemons into lemonade. The founder of TRIZ and author of the 40 principles, Genrich Altshuller, began his work in the 1940s in Baku, in the former Soviet Union, at the

time of Stalin. The government “rewarded” his activity by having him spend some years in work camps. There, Altshuller met many highly qualified experts who had been arrested during the great purges in the 1930s. He asked them to give classes and seminars. He established a “university” with one student and many professors. Thus, Altshuller obtained encyclopedic knowledge that he used to develop tools for problem solving.

- How can we amplify a harmful factor so that it becomes less harmful or becomes useful? Think about the harmful factor as a resource in a different process. Sulfur in coal or other fuel is a harmful component because the purification of the sulfur dioxide from exhaust gases requires complex technology and the unpurified gas is poisonous to people. However, if the sulfur content is increased, the production of sulfuric acid can become profitable. In the technology of flash-melting of copper sulfur is used both as fuel and raw material for the production of sulfuric acid.

During the 1940s, one problem for combat aircraft was the potential explosion of gasoline when gasoline fumes mixed with air in partially empty tanks. Carrying an inert gas to fill the empty space in the tanks would require extra weight and complexity. In that case, another resource was used the exhaust gases from the engine were produced onboard the aircraft and had much less oxygen than ordinary air. The exhaust gases were pumped into the tank and prevented the explosive mixture from forming.

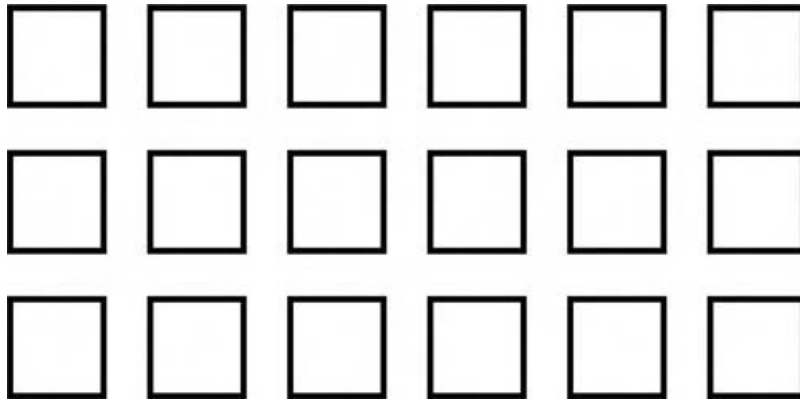
Setting backfires is a well-known technique for fighting forest fires. Controlled fires are set ahead of the fire being attacked to use up the fuel. When the fire reaches that area, it goes out.

More examples are

- Vaccination is a classic example of how to make harmful viruses to protect humans against themselves.
- Build up a tolerance to the allergen by exposure to an extract of the same allergen.
- Unhealthy salt, sodium chloride (NaCl), combined with bad-tasting potassium chloride (KCl) creates a healthy, good-tasting table salt

3.2.6. Segmentation (Fragmentation):

Transition to microlevel. Divide an object or system into independent parts. Make an object easy to disassemble. Increase the degree of fragmentation or segmentation



Segmentation. Divide a system into parts

Some examples are

- The law of the transition to microlevel, is the result of the repeated application of the principle of segmentation.
- Segmentation is a very frequently used principle. It helps to “combine the incompatible” and meet contradictory requirements in many different problems.
- More effective firefighting (good) requires more water and causes water damage (bad). Water is necessary and, at the same time, there should be no water.
- Let us segment water to small droplets and then increase the degree of fragmentation to mist. Mist can suppress fire effectively, using a very small amount of water.
- It is possible to get the idea directly, using the single principle of segmentation.

Stone washing is one of the technologies used to give denim needed features and appearance. Stones are, however, rather crude tools and clumsy machinery is needed to handle them. A nice solution is to use enzymes instead of stones to get the same result. Enzymes are not small stones, but the concept of using molecules instead of large objects comes from the principles of segmentation.

3.2.7. SEPARATION.

Separate the only necessary part (or property) or remove an interfering part or property from an object or system.

We usually need only a part of the system or some property or feature.



Separate a part or property from a system

For example:

- We need light, not lighting devices. Today, many parts of the lighting system are located some distance away from the places lighted. Reflectors and fiber optics both can separate the mechanism, such as a lamp, from the point of use of the light. Fiber optics are used in tiny surgical instruments to provide light exactly where it is needed inside the surgical area without the bulk of a lamp.
- We do not need a vacuum cleaner as such, but cleaning capacity. A central vacuum cleaning system leaves only nozzles and a piece of tubing in the apartment. Noisy and dirty parts are located where they do not disturb inhabitants.
- An electric lawnmower can work quite well if the lawn is not too large. The production of energy is removed from the lawn.
- Put a noisy air-conditioning compressor outside a building and pipe the compressed air to the place where it is needed most medical and dental offices are built this way.
- The example of fighting fire with mist illustrates the separation principle. Only small-diameter tubes are needed at the fire's location. The heavy part of the equipment is removed.
- Franchising separates the ownership of a local business, such as a restaurant or a printing shop, from the development of the concept and the systems that make it successful.